

DSA0404 – FUNDAMENTALS OF DATA SCIENCE

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1. What is Data Science? Explain its primary objective.

Data Science is the field of extracting meaningful insights and knowledge from data using statistics, programming, and machine learning.

Primary objective: To analyze data and support better decision-making and predictions.

2. Why is Data Science important in today's digital world? Mention any four reasons.

- a. Helps in better decision-making.
- b. Predicts future trends and outcomes.
- c. Automates business processes.
- d. Extracts valuable insights from large amounts of data.

3. List any four benefits of Data Science in business and society.

- a. Improves business efficiency.
- b. Enhances customer satisfaction.
- c. Detects fraud and reduces risks.
- d. Supports healthcare and scientific research.

4. Mention four real-world applications of Data Science.

- a. Healthcare (disease prediction)
- b. Finance (fraud detection)
- c. E-commerce (product recommendations)
- d. Transportation (traffic prediction)

5. What are the different facets of data? Briefly explain each.

- a. **Structured Data:** Organized in rows and columns (e.g., databases).
- b. **Semi-structured Data:** Partially organized with tags (e.g., JSON, XML).
- c. **Unstructured Data:** No fixed format (e.g., images, videos, text).

6. Differentiate between structured, semi-structured, and unstructured data with suitable examples.

Type	Description	Example
Structured	Fixed format	Excel sheet, SQL database
Semi-structured	Contains tags/labels	JSON, XML
Unstructured	No predefined format	Images, videos, emails

7. What is Big Data? State its key characteristics (5 Vs).

Big Data refers to extremely large and complex datasets that cannot be processed using traditional tools.

5 Vs:

- Volume
- Velocity
- Variety
- Veracity
- Value

8. Explain the Big Data ecosystem. Name any four technologies used in it.

The **Big Data ecosystem** is a collection of tools used for storing, processing, and analyzing large datasets.

Technologies:

- a. Hadoop
- b. Spark
- c. Hive
- d. HBase

9. What are the major phases of the Data Science process?

- a. Data Collection
- b. Data Preparation
- c. Data Analysis

- d. Model Building
- e. Model Evaluation
- f. Deployment
- g. Presentation of Results

10. What is data retrieval? Mention two common sources from which data can be retrieved.

Data retrieval is the process of collecting or accessing data from different sources.

Sources:

- a. Databases
- b. APIs (or Websites)

11. Why is data collection considered an important step in the Data Science process?

- a. Ensures accurate and reliable data.
- b. Improves model performance.
- c. Reduces errors.
- d. Supports better decision-making.

12. What is data cleansing? List any four common data-cleaning techniques.

Data cleansing is the process of correcting or removing inaccurate and incomplete data.

Techniques:

- a. Remove duplicates
- b. Handle missing values
- c. Correct inconsistent data
- d. Remove outliers

13. What is data integration? Why is it necessary in Data Science?

Data integration combines data from multiple sources into one unified dataset.

Need:

- a. Improves data quality.

- b. Provides a complete view.
- c. Eliminates redundancy.
- d. Supports accurate analysis.

14. Explain data transformation with suitable examples.

Data transformation converts data into a suitable format for analysis.

Examples:

- a. Converting text labels to numbers.
- b. Normalizing values.
- c. Changing date formats.

15. What is data analysis? Mention any four techniques used for data analysis.

Data analysis is the process of examining data to discover useful information.

Techniques:

- a. Descriptive Analysis
- b. Diagnostic Analysis
- c. Predictive Analysis
- d. Prescriptive Analysis

16. Differentiate between descriptive, predictive, and prescriptive analytics.

Analytics Purpose

Descriptive Explains what happened

Predictive Predicts what may happen

Prescriptive Recommends what should be done

17. What is meant by building a model in Data Science? Mention the basic steps involved.

Model building is the process of training an algorithm to make predictions or classifications.

Steps:

- a. Select algorithm
- b. Train model
- c. Test model
- d. Tune parameters
- e. Make predictions

18. Why is model evaluation important? Name any four evaluation metrics used for classification models.

Importance:

- a. Measures model accuracy.
- b. Detects errors.
- c. Compares models.
- d. Improves performance.

Metrics:

- e. Accuracy
- f. Precision
- g. Recall
- h. F1-Score

19. Why is presenting findings an essential part of the Data Science process? Mention any four visualization tools or techniques.

Presenting findings helps stakeholders understand insights and make informed decisions.

Tools/Techniques:

- a. Bar Chart
- b. Line Graph
- c. Pie Chart
- d. Tableau (or Power BI)

20. How are Data Science models deployed into real-world applications? Give any four examples of Data Science applications.

Models are deployed by integrating them into web, mobile, or cloud applications using APIs or software systems.

Applications:

- a. Recommendation systems (Netflix, Amazon)
- b. Fraud detection in banking
- c. Disease prediction in healthcare
- d. Spam email filtering
- e. Model Evaluation
- f. Deployment
- g. Presentation of Results

